



DOIP — Feature Catalog & Workflows (All Features, Descriptions & Flow Diagrams)



Purpose: a single reference listing every DOIP feature with a plain-language description and its end-to-end workflow diagram. This is the "what does the platform actually do" document — for the full engineering detail, each feature links to its chapter in the [Master Project Document](#). Diagrams are schematic ASCII flows.

1. Simulation Engine & Scenario Runs

What it is: the core product — a deterministic tick engine that generates the entire synthetic world: units, movement, sensors, weather, resources, vitals. Every other feature consumes its events. Same seed = identical run, which makes exercises replayable and gradeable. (*Ch. 7*)

```
Operator picks scenario + seed -> START RUN
    |
    v
Tick loop (1 Hz): clock -> scenario script -> movement
    -> weather -> logistics -> vitals
    |
    v
Every state change emitted as typed event
    |
    v
Persist (Postgres) + publish (Redis/Kafka)
    |
```

v
Consumers: map, feeds, AI, analytics ... all downstream

2. Live Map & Common Operational Picture

What it is: the home screen — a full-bleed interactive map showing every unit, incident, zone, route, and weather cell live, with severity shapes (◆△■●) and stale-data markers. Click anything → context panel. (Ch. 2, 18 — S3)

```
Engine event (position/incident/weather)
  |
  v
WebSocket push (sector-filtered) -> browser store
  |
  v
Map layer update: marker glides / pulses on arrival
  |
  v
Operator clicks marker -> context panel detail
  |
  v
Operator acts (acknowledge / task / open module)
```

3. Real-Time Event Feed, Alerts & Notifications

What it is: the chronological truth stream — every event in a filterable feed and collapsible ticker; critical events escalate to toasts, status-bar badges, and map pings. Operators acknowledge alerts; unacked counts are always visible. (Ch. 2, 6)

```
Event emitted -> severity classified (REF-02 rules)
  |
  +--> low/medium: feed + ticker only
  |
  +--> high/critical: toast + badge + map ping
    |
    v
  Operator acknowledges -> ack event logged
```

Unacked counter updates; **AAR** captures ack latency

4. Mission Planning & Route Optimization

What it is: plan unit taskings on the map — waypoints, survey zones — with A routing that costs terrain, threat zones, and weather. Naive vs optimized route always compared; launch is blocked until the feasibility check (battery, endurance, restricted zones) passes. (Ch. 2, 18 — S5)*

Planner draws waypoints/zone on map

|

v

Mission service: naive route + optimized route (**A***)

|

v

Feasibility check: endurance, battery, zones, weather

|

fail | pass

v v

reasons **LAUNCH** enabled -> tasking event -> unit follows route

listed

|

v

WaypointReached events -> progress on map

5. AI Situation Reports (Sitreps)

What it is: LLM-authored situation summaries generated on demand or on cadence — severity-ordered, ≤250 words, every claim cited to a real event id. A grounding verifier gates publication; unsupported claims are flagged, never shown as fact. (Ch. 8, 19 — prompt **sitrep-author**)

Operator hits **Generate** (or timer fires)

|

v

Event window **selected** (since last sitrep)

```

      |
      v
LLM (sitrep-author prompt) -> draft with [e:id] tags
      |
      v
Grounding verifier: every claim vs event list
      |
fail  |  pass
  v   v
flagged Published in violet AI card with citation chips
& held          |
                  v
                Chip click -> event pulses on map (proof)

```

6. Operator Q&A (RAG over SOPs)

What it is: ask doctrine and live-state questions in plain language. A router sends doctrine questions to the SOP corpus (RAG with mandatory citations) and live questions to safe SQL over world state; hybrid questions use both. "Not in SOPs" is a first-class answer. (*Ch. 8, 19 — corpus*)

Operator question

```

      |
      v
query-router: doctrine? live-state? hybrid?
      |           |
      v           v
RAG retrieve   guarded SQL over projections
(corpus chunks) |
      |           |
+-----+-----+
      |
      v
Answer with doc-id / event citations
      |
      v
No relevant chunk? -> "Not in SOPs" (never invented)

```

7. Computer Vision Detection & Review

What it is: an object-detection pipeline over prerecorded/synthetic aerial frames — vehicles, persons, smoke — with confidence scores. High-confidence detections become map events; low-confidence ones queue for human review. Every detection keeps model-version lineage. (Ch. 8, 18 — S9)

```
Footage job (synthetic frames or sourced clips)
  |
  v
Frame sampler -> YOLO-class detector -> boxes + scores
  |
  +--> score >= threshold: auto-accept
  |
  +--> below threshold: human review queue (S9)
      |
      accept | reject
      v
Detection event -> map marker + risk scoring
  |
  v
Lineage (model version) kept for debrief
```

8. Logistics Tracking & Forecasting

What it is: every depot and unit consumes fuel, medical, rations, and batteries per activity; dashboards show burn-down, and a forecasting model projects when each stock empties — with alerts before it happens. (Ch. 2, 8, 18 — S10)

```
Unit activity per tick -> consumption model
  |
  v
Stock levels updated -> burn-down series
  |
  v
Forecast (Prophet/XGBoost): time-to-empty + band
  |
  v
Threshold rules -> warning alerts ("D-2 empty ~6h")
```

|
v
Operator reallocates / tasks resupply -> new events

9. Personnel Health Monitoring

What it is: synthetic vitals (heart rate, fatigue, temperature) per person, driven by activity and circadian drift, with scripted anomalies. Threshold crossings raise alerts and suggest unit rotation. (Ch. 2, 7, 18 — S11)

Vitals system per tick: baseline + activity + drift
|
v
Per-person series -> roster grid (ok/amber/red)
|
v
Threshold crossing -> alert + map badge + ticker
|
v
Operator opens detail -> trend sparkline + suggestion
|
v
Rotation tasking (optionally via War-Room task board)

10. Weather Simulation & Operational Effects

What it is: weather cells (rain, storm, fog) that spawn, drift, and evolve — and *matter*: they slow units, ground UAVs, raise route costs, and degrade sensor reports. Weather is a cost layer in every planning decision. (Ch. 7)

Scenario spawns cell (or rule triggers one)
|
v
Cell drifts/evolves per tick -> map overlay updates
|
v
Units inside cell: speed multipliers applied
|

```

      v
Mission service: weather cost layer in routing
      |
      v
Sensor/report noise increases in affected area

```

11. Replay & After-Action Debrief

What it is: any run can be replayed from its event log alone — scrub, jump to events, watch decisions unfold at 1–8x. The after-action report auto-builds the timeline, metrics, decision review, and lessons. (*Ch. 7, 11, 18 — S6*)

```

Completed run (events in DB)
      |
      v
Replay reader streams events between t1..t2
      |
      v
Map renders historical state at scrub position
      |
      v
Annotations / injects / decisions appear at true sim-time
      |
      v
AAR export: timeline + metrics + decision review (POL-03)

```

12. Analytics & KPIs

What it is: per-run and cross-run metrics — p95 response time, incidents-within-target, resource efficiency, comms-loss minutes — plus run-vs-run comparison on the same scenario and seed to measure team improvement. (*Ch. 11, 18 — S12*)

```

Events -> metric aggregation jobs
      |
      v
KPI store: per-run scores + distributions
      |

```

```

      v
Dashboards: histograms, timelines, efficiency
      |
      v
Compare mode: RUN-A vs RUN-B (same seed) deltas
      |
      v
Feeds Ch. 14 gap attribution + AAR exports

```

13. What-If Theater — AI Scenarios & Animated Playback

What it is: describe a hypothetical in plain language ("storm hits sector C during the convoy run") — the AI drafts scenario actions in a validated DSL, a human approves, the engine runs it headless fast, and the result plays back as an animated map story with cited AI narration. The AI authors *scenarios*, never outcomes. (Ch. 13, 18 — S8)

```

Natural-language what-if prompt
      |
      v
scenario-author LLM -> whitelisted DSL actions (JSON)
      |
      v
Validation Gate -> human review [edit/approve/discard]
      |
      v
Headless fast run (engine computes outcomes)
      |
      v
Animated condensed playback + cited narration
      |
      v
Compare vs baseline run

```

14. Preparedness Investment Advisor

What it is: import a (non-classified) budget and inventory, run a scenario battery against it, get a ranked capability-gap register with run-level evidence, then budget-constrained acquisition portfolios — each validated by re-simulation on the same seeds and benchmarked against documented real deployments. (Ch. 14, 18 — S13)

```
Budget + inventory import (CSV/JSON, validated)
      |
      v
Scenario battery (10-30 seeded runs)
      |
      v
Gap attribution -> ranked Gap Register + evidence links
      |
      v
MILP optimizer -> 2-3 portfolios within budget
      |
      v
Re-simulation (same seeds) -> before/after deltas
      |
      v
Historical parallels panel (cited, fact vs projection)
      |
      v
Report export with decision-support disclaimer
```

15. War-Room Mode — Multi-Operator Sessions

What it is: several people run one exercise together — Commander sees everything and owns the task board; Sector Operators get server-filtered sector views; Observers watch silently. Annotations and taskings are events, so the whole team session replays in the debrief. (Ch. 15, 18 — S14)

```
Commander creates session -> role assignment
      |
      v
Operators join -> sector-scoped WS streams
      |
```

```

      v
Shared map: live annotations + named cursors (4 Hz)
      |
      v
Commander assigns tasks -> ack/doing/done states
      |
      v
All collaboration persisted as events
      |
      v
Team debrief: full session replay + per-role timing

```

16. Red-Team Agent — Adaptive Training Adversary

What it is: an AI adversary that watches team performance and injects fair, bounded complications (diversions, comms degradation, resource pressure) from a whitelisted playbook when the exercise is too easy. Every inject is disclosed in the debrief — mechanics come from the playbook, never free-form AI. (*Ch. 16*)

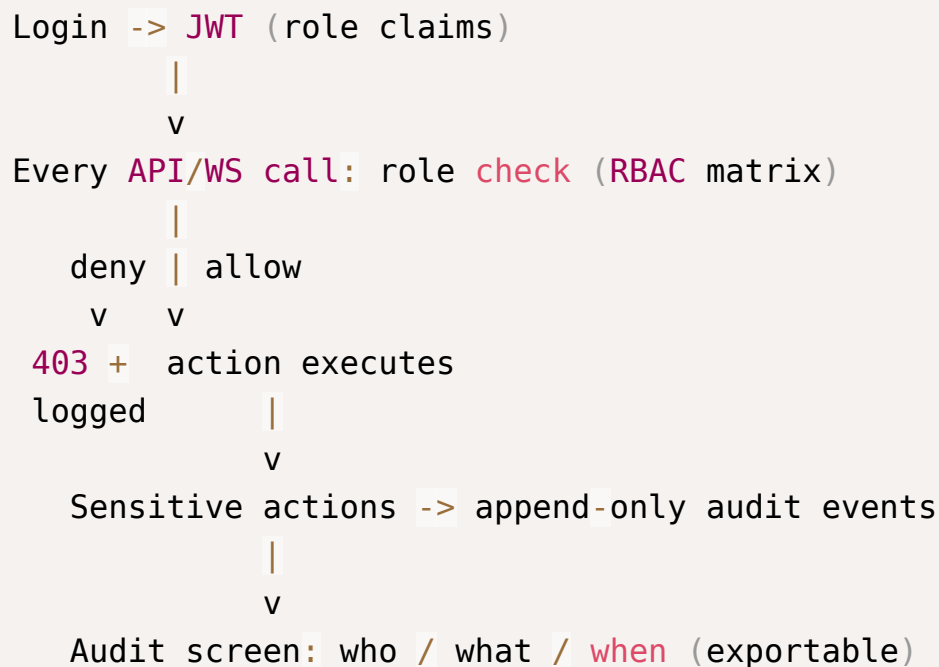
```

Live run metrics -> performance assessor
      |
      v
Director policy: too easy? too hard? fair window?
      |
      v
Inject planner picks playbook entry (bounded)
      |
      v
Validation Gate (same as Ch. 13) -> InjectApplied event
      |
      v
Team responds (they see effects, not the hand)
      |
      v
Debrief: every inject revealed with timing + rationale

```

17. Access Control, Roles & Audit

What it is: JWT auth with four platform roles (admin / planner / operator / viewer) plus session roles in War-Room mode; every sensitive action — imports, scenario edits, purges — lands in an append-only audit log. (*Ch. 10, 18 — S15*)



Reading order for implementation: features 1–3 are Phase 1 (the demoable core), 4–6 Phases 2–3, 7–10 Phases 4–5, 11–12 Phase 6, 13–16 post-v1 extensions, 17 spans all phases. Full sprint mapping lives in Ch. 9 of the [Master Project Document](#).